

Qubit Evolution

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Artificial Intelligence and Computer Science
A Deep Learning Project

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Problem & Tasks

Goal: learn temporal dynamics and generate missing/unknown timesteps.

One pipeline, two tasks

Input → **Model** → **Output**

Forecasting (FC)

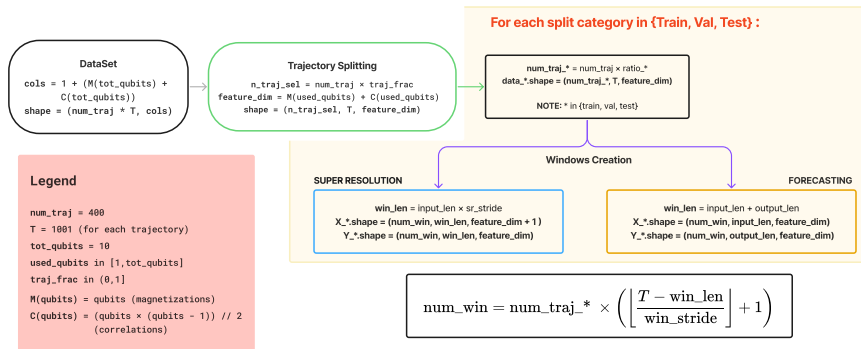
previous steps → future steps

Super-Resolution / Imputation (SR)

gapped window → filled window

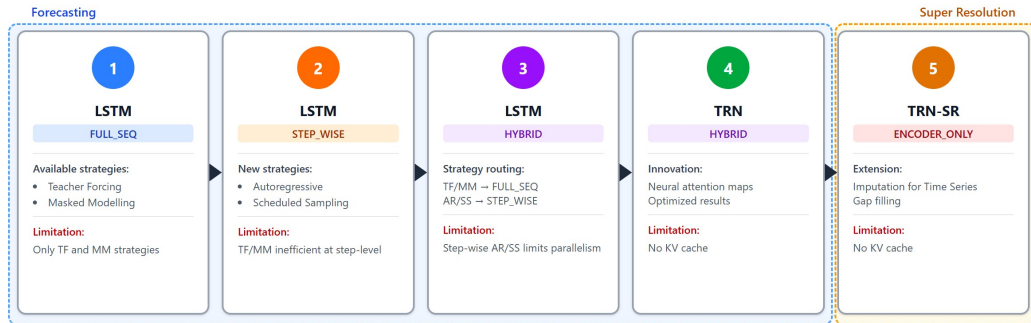
Dataset & Preprocessing

Same raw data, **different preprocessing** for FC vs SR.



Note: SR adds a mask channel to distinguish observed vs missing samples.

Model Roadmap



Abbreviations: TF = Teacher Forcing; MM = Masked Modelling; AR = Autoregressive; SS = Scheduled Sampling

Decoder mode	TF / MM	AR / SS
FULL_SEQ	parallel	–
STEP_WISE	step-wise	step-wise
HYBRID	parallel	step-wise

Curriculum & Prediction Mode (Forecasting)

Curriculum:

Example (output_seq_len = 80): $H_k = c_k \cdot 80$, where H_k is the horizon at phase k .

Phase k	Curriculum c_k	Loss range
1	0.4	[1 : 32]
2	0.8	[1 : 64]
3	1.0	[1 : 80]

In all, we still predict 80 steps each phase; in horizon, we predict only H_k .

Prediction Mode:

Mode	Pred length	Loss on	Notes
all	output_seq_len	[1 : H_k]	stable graph, more compute
horizon	H_k	[1 : H_k]	less compute, may re-trace

Free-Running Evaluation

Why: training loss is optimistic (assisted inputs)

Probes

Probe	When	Purpose
fr_phase	every 2 epochs	feedback at current horizon
fr_target	end of phase	full-horizon performance
fr_curve	end of phase	error vs horizon

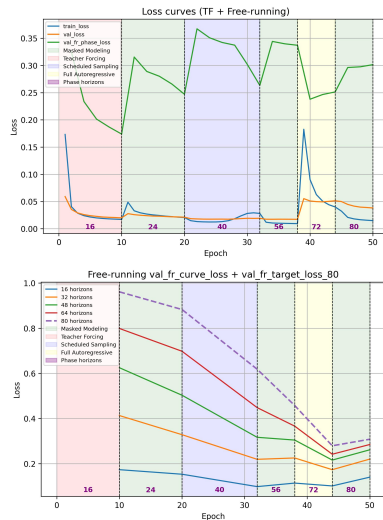


Figure 1: Loss on TRN Hybrid.

Hyperparameter Tuning (Optuna)

Two-stage optimization:

Level 1: model-capacity hyperparams

Level 2: windowing hyperparams
(input_seq_len, output_seq_len,
stride)

Trial ranking (scalar score)

$$\text{Score} = 0.70 L_{\text{fr_target}} + 0.25 L_{\text{fr_curve}} + 0.05 L_{\text{fr_phase}}$$

$$L_* = \text{loss}$$

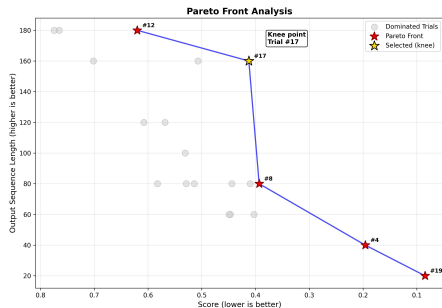


Figure 2: Pareto Front of LSTM Hybrid

Objective	Direction
Score	minimize (↓)
Output length	maximize (↑)

Table 1: Level-2 multi-objective

Results — Forecasting

Takeaway: TRN follows the target longer

Fixed windowing: $(L_{in}, L_{out}, s) = (200, 50, 20)$; Level-1 best per model; 50 epochs on Tesla T4.

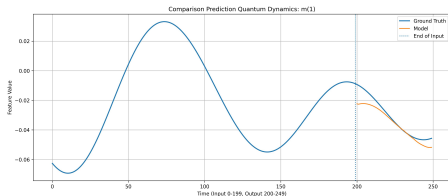


Figure 3: LSTM Hybrid

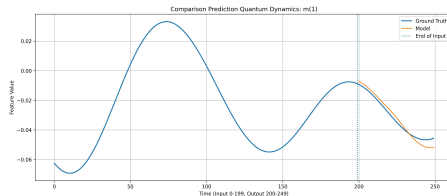
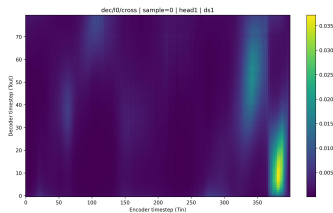


Figure 4: TRN Hybrid

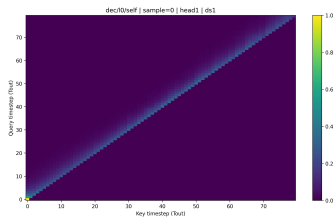
Model	FR_target ↓	MSE ↓
LSTM Full-seq	0.7170	0.0081
LSTM Step-wise	0.2266	0.1252
LSTM Hybrid	0.2510	0.1720
TRN Hybrid	0.2444	0.0145

Results — Transformer Attention Maps

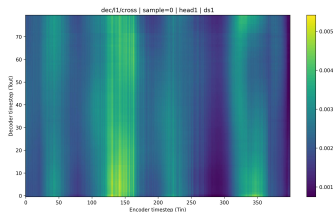
TRN Hybrid knee-point configuration; decoder head 1.



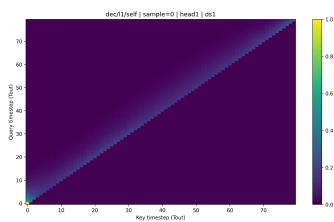
dec/10/cross



dec/10/self



dec/11/cross



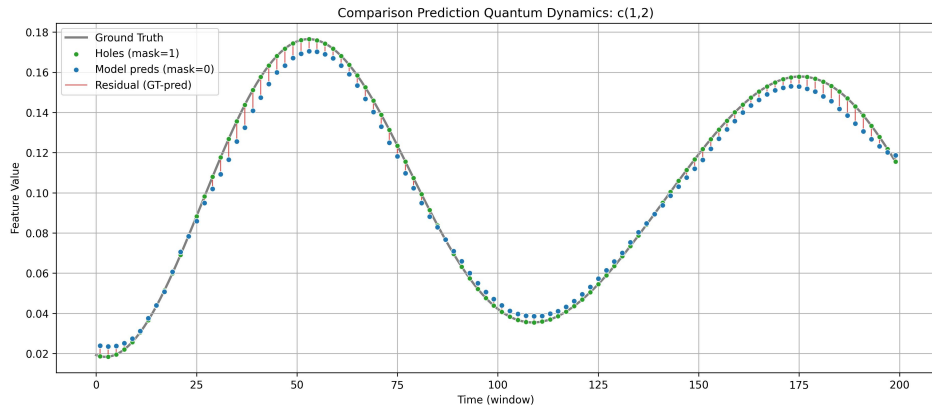
dec/11/self

Results — Super-Resolution (Imputation)

Takeaway: good qualitative reconstruction; larger errors near sharp transitions.

Trained 50 epochs on Tesla T4.

Residuals: $r_t = y_t - \hat{y}_t$ (on masked points).



Limitations & Future Work

Today

- step-wise Scheduled Sampling (SS)

Limitation

- limited parallelism $\Rightarrow$ slow for long horizons



Future Work

- two-pass SS (Transformer, more parallel)
- 3D Pareto: ($\downarrow$ Score, $\uparrow Len_{out}$, $\downarrow Len_{in}$)
- alternatives to reduce exposure bias

Thank you!

Questions?

Source code and artifacts are available on GitHub:

[qubit-evolution-dl](#)

